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COMPUTER SCIENCE & ENGINEERING
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# EXPERIMENT–3

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## Aim

To design responsive and reusable user interfaces using modern React UI libraries.

## Objectives

- Design responsive dashboards using Material UI.
- Implement complex tables using Ant Design.
- Create reusable styled components.

## Easy Level Program – Material UI Responsive Dashboard

```
import { Grid, Paper } from "@mui/material";

function Dashboard() {
  return (
     <Grid container spacing={2}>
       <Grid item xs={12} md={6}>
         <Paper style={{ padding: 20 }}>Card 1</Paper>
       </Grid>
       <Grid item xs={12} md={6}>
         <Paper style={{ padding: 20 }}>Card 2</Paper>
       </Grid>
     </Grid>
  );
}

export default Dashboard;
```

### Output

Responsive dashboard layout rendered successfully. Cards adjust automatically based on screen size.

## Medium Level Program – Ant Design Table with Sorting and Filtering

```
import { Table } from "antd";

const columns = [
  {
    title: "Name",
    dataIndex: "name",
    sorter: (a, b) => a.name.localeCompare(b.name),
  },
  {
    title: "Age",
    dataIndex: "age",
    filters: [
      { text: "20s", value: 20 },
      { text: "30s", value: 30 },
    ],
    onFilter: (value, record) => record.age >= value,
  },
];

const data = [
  { key: 1, name: "John", age: 28 },
  { key: 2, name: "Alice", age: 32 },
];

function UserTable() {
  return <Table columns={columns} dataSource={data} />;
}

export default UserTable;
```

### Output

Data table displayed with sorting and filtering options. User can sort by name and filter by age.

## Hard Level Program – Styled Components Reusable Button

```
import styled from "styled-components";
```

```
const StyledButton = styled.button`
  background-color: #2962ff;
  color: white;
  padding: 10px 20px;
  border: none;
  border-radius: 6px;
`;

function App() {
  return <StyledButton>Click Me</StyledButton>;
}

export default App;
```

### Output

Reusable styled button rendered. Consistent styling applied across application.

## Learning Outcomes

- Learned how to design responsive layouts using Material UI.
- Implemented sorting and filtering features using Ant Design tables.
- Created reusable UI components using Styled Components.
- Improved UI scalability and maintainability in React applications.